

Towards Environmental Justice in Transportation Decision Making using Structured Public Involvement

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ABSTRACT

Environmental Justice (EJ), in the form of distributional justice, is mandated by a 1994 Executive Order. However, EJ is not easily achieved. This article examines how a philosophy based on John Rawls' theories of procedural justice and access to justice can address the need to achieve distributional justice (EJ).

The DoT's definition of EJ is used to explain how the philosophy of large group processes can address the requirements. EJ research can be divided into identification and mitigation strategies. EJ mitigation strategies intersect with public involvement, which in transportation has a long, and often controversial, history. To improve procedural justice, the authors examine how effective large-group processes might deliver high-performance public involvement. To realize this contribution, methodological barriers in group performance are discussed and the role of technologies such as electronic polling and visualization is discussed. The authors propose four process metrics for public involvement. Using Structured Public Involvement or SPI project data, the authors argue that designing processes that satisfy these metrics will help to improve procedural justice and thereby address specific EJ aims. Data from this research and the cross-disciplinary literature review illustrates that realizing this potential improvement will require; a philosophical shift to a higher Arnstein Ladder level; the identification and use of stronger mixtures of methodologies for involving large groups of citizens with diverse values; and the integration of these valuations into effective decision support systems for project managers and engineers.

Transportation infrastructure is a public good, in the sense that public monies are invested in planning, design and construction and maintenance and replacement of infrastructure and facilities (\$221.7 billion in total in 2007 according to the Bureau of Transportation Statistics [47], or over \$700 per person per year in the United States if normalized). The question of distributional impacts i.e. the spatial concentration of benefits and costs and the relationship to specific social, economic and demographic groups, of public goods investment and management, particularly transportation, has long been considered a problem [2]. Since the 1960s, the relationships between transportation, the environment, and equity and justice began to be apparent. Research began to investigate these topics across transportation system dimensions such policy [3,4], traffic flow [5], eminent domain [6], parking [7], and numerous others. In the early 1980s, the term environmental justice was defined to formalize the relationship between disadvantaged social groups and their environment with reference to hazards such as waste [8]. It became evident that all public goods investment and management shared similar distributional problems and the concept of environmental justice began to appear in transportation analyses [9]. Observation and documentation yielded to a focus on mitigation in the 1990s.

Then, in 1994, Environmental Justice (EJ) was formalized by Executive Order 12898 [10].

According to the U.S. Department of Transportation [11], the aims of EJ are:

“1) To avoid, minimize, or mitigate disproportionately high and adverse human health and environmental effects, including social and economic effects, on minority populations and low income populations. 2) To ensure the full and fair participation by all potentially affected communities in the transportation decision-making process. 3) To prevent the denial of, reduction in, or significant delay in the receipt of benefits by minority and low-income populations.”

Martin Wachs [12, p.151] believes that “EJ issues arise most frequently in five contexts,” including the disproportionate impact or benefit of transportation projects on different populations, and in relation to the taxes that they pay. The larger the project, the more problematic these become, due to the spatial concentration of impacts [13]. In the transportation domain, where capital expenditures are large, and where projects span large geographical scales and extend over long timeframes, ensuring that such investments are EJ compliant has assumed large dimensions [14].

Functionally, EJ processes are divided into identification and mitigation, meaning first documenting an EJ problem, and then developing and applying methods to address the DoT’s stated EJ aims in real-world situations. To identify EJ issues, new approaches are being tested, for example geospatial analytic methods [15,16]. More recently, increasing attention is being paid to the broader EJ context of societal relevance to all stakeholders, e.g. its relationship with “social sustainability” [17], planning methods [18] and urban modeling [19].

However, much of this research is aimed largely at goals (1) and (3), that is, the distribution of benefits and disbenefits, or what philosopher John Rawls (and others) refer to as ‘distributional justice.’ [20] Such analyses give themselves more readily to quantitative and spatial measurement, which facilitates more conceptually direct comparisons between population subgroups, such as designated EJ populations.

Less thoroughly and rigorously analyzed is goal (2), what Rawls characterizes as ‘procedural justice’ (PJ) and ‘access to justice.’ Oddly, the proportion of research devoted to PJ techniques, particularly in transportation, is much more modest, and sometimes consists of anecdotal case studies with little systematic or comparative analysis of processes or their relationship to goals and outcomes. Existing EJ efforts often focus on targeted outreach, recruitment and participatory activities involving specific, under-represented groups such as low-income groups [21], women [22] and/or ethnic minorities [23-26] and socially disadvantaged groups [27]. For example, the FHWA website for EJ provides “Best Practice” case study narratives, along with a list of “Public Involvement Tools.” However, it references interested parties to the National Highway Institute’s public involvement training courses if they desire to learn more about how to improve the PJ aspects of their EJ projects [28]. This may be appropriate in that existing mandates for public involvement under NEPA [29], ISTEA [30] and SAFETEA-LU [31] frequently may involve the same stakeholders as EJ processes [32]. However, there are no formal standards against which to evaluate the comparative efficacy of various EJ processes e.g. [33-36].

Relationship between EJ and Public Involvement

Public involvement in transportation decision-making has a relatively long, and often controversial, history [37]. Public involvement in Federal expenditures and policy was first formalized by the National Environmental Policy Act of 1970 [29]. Many transportation infrastructure projects, because of their scale and impacts, involve Environmental Impact Statements (EIS) and other processes that mandate public involvement [38]. As transportation agencies integrated these mandates into the overarching structures provided by ISTEA in 1991 [18] and then SAFETEA-LU in 2005 [19], various sets of guidelines and principles have been published by Federal agencies such as the Federal Highways Administration in an effort to clarify and codify public involvement in the transportation sector [28,39]. However, these mandates do not specify the form, content or objectives of this public involvement and there are, at present, no formal process performance metrics. Further, EJ compliance requires the involvement of citizens, sometimes in appreciable numbers; however the relationship of EJ processes to public involvement more broadly is often not clear either to agencies or citizens.

Lessons from Group Process Research in Cognate Fields

Fortunately, transportation is not alone in its need for better understanding of public processes across all scales and groups. Many other fields of public policy and public goods management face similar challenges. Rowe and Frewer published a highly-cited article on evaluating participation in a public policy journal [40]. Environmental management literature offers a range of publications dealing with quality and outcomes for group processes e.g. [41]. Group systems literature also presents a number of metrics for process evaluation [cite]. In the field of health care alone entire journals focus on analyzing the problems associated with public input into health care regimens and treatments, with research delving into even the comparative properties of different data gathering methods for different size groups [e.g. 42-44]. Civil engineers working in water policy, with training and professional backgrounds similar to those in transportation, are also now deeply involved in more productive ways to incorporate and measure the quality of public participation in their decision-making as concerns water infrastructure. For example, the US Army Corps of Engineers offers extensive and detailed

online resources, models, and methods for the collaborative decision making on their Shared Vision Planning website [45]. They have even fostered the development of technical models of water resource systems whose inputs and outputs can be tuned based on the information gained from the users: i.e. the public. This concept is transferable to transportation planning, for example, local residents could work with transportation professionals to define what the true ‘level of service’ is on their roads, and what would count as ‘improvement’ prior to developing plans. This is relevant because professionals in most of these fields are subject to the same practical and philosophical EJ challenges as transportation professionals.

In addition, there are more recent political calls for increased accountability, such as the recent Open Government Directive [46] and the Champions of Participation report generated from a deliberative process among all Federal agencies, including transportation agencies [47]. Consequently, the quality of public involvement processes in transportation, and especially for EJ processes, deserves more attention. Critically understanding how citizens, regardless of how they are grouped or configured, can be more effectively integrated into public goods decisions processes is a parallel challenge that confronts EJ no less than it confronts public involvement writ more broadly.

The authors address these challenges in this article. We argue that EJ goals will be more closely approached when high quality public processes are used systemwide and not only with targeted “EJ” populations. The procedural justice aspects called for in EJ need not be considered merely a parallel process entailing additional inconvenience and resource commitments from the agency and citizens: this process can contribute materially to the distributive justice aspects of EJ. Further, the lessons learned from literatures about better procedural justice processes for public involvement generally are useful and applicable to the ongoing EJ challenge as well. The authors argue that better distributive justice can be achieved by delivering higher quality procedural justice, including processes that seek to include the maximal number of people of all backgrounds, and whose quality is measured objectively using transparent, stakeholder-driven indicators. Four process performance criteria, Quality, Inclusion, Clarity and Efficiency, will be identified and discussed for their relevancy to the challenge of procedural justice in EJ.

Why Procedural Justice Matters

The philosophy of our argument is premised on the assumption that absolute distributive justice, i.e. the truly equitable spatial and social distribution of costs and benefits as the focal point of EJ research and work, is a worthy yet unattainable goal. However, it can be improved by strengthening procedural justice processes across the entire population [48, 49, 50]. Improved PJ is linked to the distributional aims of EJ for these three reasons:

1. *High quality procedural justice increases participation by the disenfranchised (i.e. EJ populations) and thus the quality of data from the public that informs the project.* Numerous studies in a variety of public processes and environmental fields have shown that participants make judgments about public participation based on a comparison of effort required versus efficacy of input [51]. Populations that have experienced less effectiveness over time are less likely to participate in future processes [52-56]. Consequently, conventional EJ processes compensate by targeting these presumed

populations. However, there is no method for assuring that all such disenfranchised populations have been identified and reached when the determination of their existence is expert-driven. It is not clear, for example, that census-driven ethnicity targeting can fulfil EJ aims when even membership in these ethnic categories is not clear to respondents themselves.

2. *Publicly-sourced information and judgment augments existing expert-driven distributive justice analyses.* To the extent that disenfranchised groups are not reached, the information available to professionals is compromised. The problem of allowing only technical expert analyses of projects is well-known in EJ research [57,58]. Better procedural justice allows for more accurate identification and measurement of benefits and disbenefits from the point of view of the EJ and overall community. Many of the issues surrounding project trade-offs are not susceptible to cost-benefit analysis [59]. Instead, they must rely on ethical and other judgments that emerge from the value systems of the subject populations. This requires that the affected populations be fully identified and that their judgments be accurately and thoroughly solicited. But this problem can only be addressed through better procedural justice (i.e. improved process).
3. *Reliance on segregated small-scale processes during EJ analyses can exacerbate interest group politics, and undermine the original aim of legitimacy* [60]. Pre-ordained groups can quickly become arrayed against one another, with the process degenerating into a battle over which group is more privileged and can assert more strongly its “solution” [61, p.703]. Broad-based data, conversely, can offer stronger cross-group legitimacy [62, p.63].

For these reasons, the goal of improving procedural justice in public goods questions deserves more attention. However, the 2000 TRB White Paper on Public Involvement notes that the problem begins with the lack of clarity regarding what constitutes “good” or “high performance” public involvement [63]. In an article titled “Why Public Involvement is Made So Ineffective,” Connelly examined numerous processes for public involvement [64]. Some findings include; unclear objectives for public meetings; a lack of good faith and transparency in the conduct of such meetings; and a lack of integration of the output data, if any, with consultant technical and engineering decision making. In the following section, we will discuss some of the challenges to high quality public process and their implications for EJ goals, particularly participation.

Challenges to Procedural Justice in EJ Processes

Some Citizens are More Equal Than Others

In public involvement in transportation, as in other public goods fields, a system has evolved in which entities that possess legal resources and training can indulge in gaming behavior to achieve disproportionate influence on the project and its outcomes. For example, the solicitation of groups to sign on as “consulting parties” under the NEPA process has the effect of creating “tiered citizenship.” Such entities have disproportionate access to ongoing project information and may be invited to certain types of project meetings, allowing them to affect processes in ways not available to others. Such entities may also exhibit influence through the assertion of representation “on behalf of” other citizens at public meetings. Even higher levels of influence may be granted to certain citizens through the creation of advisory panels [65]. This is not

helpful to the project team, nor to other citizens. This failure to foster civil, equitable and focused dialog with the majority of stakeholders contributes to a phenomenon that some media term “civic detachment” [66], or what planners term “the rational ignorance” of these processes on the part of the citizens [67]. This may be especially true of the most disenfranchised citizens, which would include some EJ populations.

As a consequence, the execution of transportation projects yields four classes of citizens: advisory panels, consulting parties, and EJ populations, and everyone else (“the public”). Further, the projects typically specify different types of engagement for each of these groups, further exacerbating the procedural injustices across the population. An express goal of identifying EJ populations and working with them separately is to attempt to offset some of the inherent advantages possessed by politically powerful parties. While this may well be necessary, another method of reducing procedural injustice to EJ populations is to flatten the power distribution, that is, to reduce the procedural injustice to *all* populations by working harder to provide equal access to project input to all participants, while simultaneously minimizing the role of, and need for, ‘special’ groups like advisory panels and consulting parties.

The Wisdom of Crowds.

Conventional public involvement generates fewer problems when conducted at small scale meetings, and/or with people of homogeneous training, cultural backgrounds, skills and value systems [40]. However, flattening the power distribution requires the involvement of the maximum number of people and providing each of them with the same quantity and quality of input methods. It is sometimes assumed by project managers that the problems of conflict, lack of participation quality, and lack of data gathering efficiency all scale up in proportion to the number of participants. For instance, if a public meeting with twenty is difficult to manage, a meeting with two hundred is assumed to be at least ten times more challenging. For many transportation professionals, large group processes implies sponsoring something akin to improvisational theater or “karaoke night,” where a few vocal participants dominate the proceeding and very little in the way of useful information is gained. This belief can lead to counterproductive behavior; for example, attempting to discourage attendance or to control who participates at mandated public meetings [68,69]. However, what the TRB White Paper terms “overweighting of the voice of activists” [63, p.5] is not the inevitable result of eliciting data at large, open public meetings. This classic imbalance is a methodological failing. It is the result of a process that does not ensure all participants’ values are elicited openly and recorded equally and transparently.

Identification of ‘Groups’ and Their ‘Interests’

The current state of practice for EJ populations lends itself to small group processes centered around the identified community(s). Participants from the identified community(s) are solicited for the needed project input information using standard small group processes, and then these interests are incorporated back into the overall project process. While such an approach is certainly useful for the qualitative information it provides, it may actually compromise on the intent of the EJ directive by limiting who is allowed to be classified as ‘EJ’ and thus excluding others who are ‘near EJ’ or even ‘don’t know we’re in EJ.’ That is, the expert definition of an EJ population is hardly a tidy operation, yet project managers are left with only two process membership categories to work with: EJ or not.

This forced choice, combined with small group processes, is unhelpful to the EJ community, the project management and citizens at large. Further, it is unnecessary. Tools and techniques exist that can allow project managers, in open public meetings, to gather much more nuanced information about the characteristics of the participants, including factors related to potential EJ classification, while preserving the anonymity of the participants.

Figure 2. Dr. Grossardt conducting a public meeting during an SPI protocol for transit-oriented development. Louisville, KY 2002.



The Ideology of Consensus

Methodologically, the ideology of consensus in the field of public involvement is one of the most significant impediments to attaining higher performance. Consensus has become a normative and exclusive organizing philosophy for much of the training that public involvement practitioners receive. The 2000 TRB White Paper on Public Involvement, for example, states that “One important objective of a good public involvement process is the extent to which the process builds consensus on the path to decision” [63, p.2], although this is then qualified using the terminology “informed consent.” Consensus seeking processes may work well in small group settings with participants who possess homogeneous beliefs, values and even training, and who have a stake in reaching such consensus [70]. However, attempting to reach consensus is impossible in real world situations when dealing with TI projects, particularly large scale

projects involving many hundreds, or thousands, of citizens, all with diverse preferences, tastes, training and values. Designing a process to achieve “consensus” under these conditions entails forcible participant exclusion, or the silencing of opinions outside the normatively acceptable. This is achieved overtly, for example by making it difficult or impossible for others to speak at meetings, or covertly, by not inviting dissenting or “problematic” individuals to be part of focus groups or discussion teams, or by scheduling meetings at times and places to which only certain, preferred, people can respond. As Beierle and Cayford’s review of 239 different cases, attaining consensus was often achieved “by leaving out participants or ignoring issues” [54, p. 48]. There is even literature directed at managing public involvement in this way [71]. In some cases, the lack of agreement is accommodated through such strategies as including a “Consensus with Dissent Noted” (or similar) document [72]. It can also be achieved by project managers or consultants telling public participants that their opinions are inappropriate or unfounded, or by using engineering and technical experts to enforce what participatory techniques researcher Kyem [73] terms “selective competency requirements for participation.” This approach rests on the widespread and commonly expressed view among professionals that the public is not technically qualified to advise them [74]. It is even achieved by editors and reviewers preventing the expression of the view that consensus can be counterproductive for high-performance public involvement in transportation [75].

What implication does this have for EJ processes? The hazards posed by the drive to find, or worse construct, consensus apply to communities that are particularly adversely affected by a public works project. The quality of project outcomes can only be improved by project management making every effort to assure themselves that they, and the participants, thoroughly understand the full range of values and preferences of the community that is being variably impacted. Devising the solution that is most responsive to this complexity is not synonymous with pushing for everyone to agree that the same solution is the best one for everyone and silencing, excluding or ignoring those who do not.

Examples of Processes That Address Procedural Justice Issues for EJ

At this point it should become apparent that the goals of distributional and procedural justice for EJ communities (and everyone else) are not contradictory: rather they are complimentary. Addressing them requires what systems theorist Checkland terms a soft-systems approach [76], that uses an analytic framework to integrate professional roles and responsibilities and public valuations in ways that minimize role conflict and maximize attainment of procedural justice.

One way of doing so is illustrated in the Structured Public Involvement protocols [77]. While this paper is not intended to promote SPI as a single best solution, the use of SPI on a wide variety of public goods problems has provided useful observations and large process performance data sets that bear on the issues delineated above.

SPI protocols are used to create large-scale group processes that accommodate consideration of both open-ended (i.e. brainstorming or issue probing questions) and defined scenarios (alternative outcomes) and produce data and/or quantitative models to provide decision support for project managers. SPI seeks to reduce controversy and to increase public satisfaction with the infrastructure design process and product by addressing procedural justice goals. SPI does

not mean public control; it means gathering large group input in a structured way using advanced analytic methods that allow designers access to these preferences. This protocol also allows the collection and analysis of data that supports the identification of potential EJ populations and other differentially affected populations, and the identification of their values and preferences, all in a large group setting.

SPI protocols combine the use of Audience Response Systems (ARS) for feedback with representation/visualization tools to aid in comprehension and useful feedback, and use the resulting data to create information relevant to the question at hand (Fig. A). In the instance of a physical planning or design problem, these require delineating a feasible option envelope, within which solutions are technically, financially, legally and ethically feasible. The public involvement theorists work with the project sponsors and contractors to exclude infeasible options prior to the public evaluation phase. When the full, feasible range of options is identified, the team then develops representations, such as geovisualizations or animations, to use as data elicitation methods with large groups of stakeholders. Electronic polling allows the evaluations to be conducted simultaneously, independently, anonymously, and rapidly. The project team and its members do not manipulate or coerce the participants toward specific options, as such actions would contaminate the validity of the data. The public sees the SPI method at work, and is then invited to evaluate the process. The effective partitioning of the decision envelope provides higher quality process and data for citizens and project management. This is one of the reasons for the high documented process satisfaction of SPI e.g. [79-86].

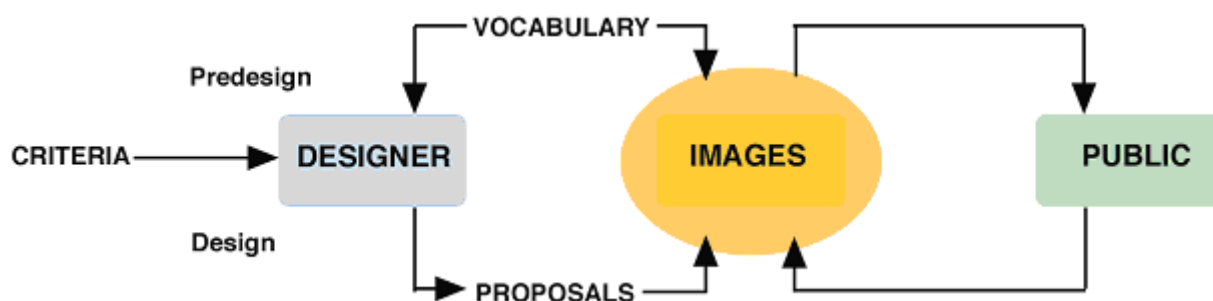


FIGURE 1A. SCHEMATIC OF RELATIONSHIP BETWEEN DESIGNER AND PUBLIC USING SPI PROTOCOL FOR A DESIGN PROBLEM

Because the ARS systems gather data rapidly and anonymously, participants can provide a great deal of useful information in a very short time. The ARS system also tracks each unique keypad, so that demographic information such as age, race, income, and even geographic location can be gathered and cross-tabbed with project data gathered in the same setting. Thus identifying the presence and preferences of subgroups such as potential EJ populations need not be limited to a technical process of identification and segregation. While such processes can still be pursued through small group meetings, any interested party from an EJ community can come to an open public meeting and contribute her ideas and preferences, anonymously and on an equal footing with all other citizens. Indeed, the process of identifying EJ communities could potentially be improved, through open meetings that allow community participants to anonymously self-identify along scales that are traditionally used to designate EJ communities. Such sub-group

demographic data has already been collected and used by the authors and their agency partners in, for example, public transit design projects [87], bridge design projects [88], and nuclear plant remediation [89].

PROCESS EVALUATION METRICS

So far this discussion has argued that the value of good procedural justice processes (part of the EJ executive order) also serve the more intuitively obvious but operationally challenging distributive justice goals of the EJ executive order [11]. On the basis of this review, and others e.g. [90], the authors argue that much of the published work that concerns procedural justice is less systematic than the large body of research that addresses EJ distributive justice goals. Such a lacuna is unfortunate and unnecessary.

Evaluation both of public processes and EJ outcomes is one of the most critical and pressing issues confronting the transportation community. EJ evaluation is problematic e.g. [91, 92, 93, 94, 95]. Public involvement evaluation is almost absent. Transportation lags other fields in this respect [96]. The TRB White Paper on Public Involvement does not contain specific proposals for evaluation metrics [63]. In a 1999 document the Transportation Research Board proposed thirty two metrics, however all were proposed to be measured by the professionals charged with delivering the processes! [97]. This is clearly inconsistent with the broader democratic goals espoused by transportation planners, design coalitions and public agencies such as FHWA [98]. In the transportation sector at the state and local scales, when such evaluation of public involvement has been conducted, it is often done by the professionals or their clients, using informal criteria or anecdotal observation, and is not subject to independent, or more appropriately, stakeholder analysis [99]. Recent research into this, commissioned by State Departments of Transportation, [100] also does not include critical stakeholder-driven metrics. The USDOT recognizes the link between public involvement and EJ [101], highlighting “effective public involvement tools for promoting EJ principles.” However, very little data exists to support the “effectiveness” claim. For these reasons, the authors offer the following proposal.

FOUR PROCESS PERFORMANCE CRITERIA

The authors propose four major themes to measure the procedural justice performance qualities of public processes. These themes draw from TRB’s 1999 evaluation document [97] and the Florida Department of Transportation’s recent evaluation of process quality indicators [100], but they also include a synopsis of literatures outside transportation, including public administration, environmental management and group systems.

These themes are Quality; Inclusion; Clarity; and Efficiency. Table 1 presents these themes, together with a number of associated metrics and indicators.

Table 1. Performance Measurements for Public Processes

Criterion	Indicators	Data
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Process Quality (Q)	Participant satisfaction	Process quality evaluation conducted anonymously by stakeholders at open public meetings
Inclusion (I)	Number of organizations, citizens and groups	Count and document attendees, participant groups during process, access to information, relevance of information etc.
Clarity/utility of decision support (C)	Expert evaluation	Testimonials, narratives, comparisons to state of the art methods for decision support
Efficiency (E)	Cost and time	\$ spent on public involvement, time taken and demanded, in relationship to performance achieved (e.g. C, I, and Q-metrics)

For stakeholder-driven Q-metrics very little data exists. Table 3 summarizes process satisfaction data gathered by the authors over the past ten years, in open public meetings using anonymous ARS systems, under various Structured Public Involvement protocols. Detailed descriptions of these various projects can be found in project-specific results published by TRB e.g. for noisewall design [82], for large context-sensitive bridge design [83], transportation/land use planning [84], and in other forums [85,86]. Similar Q-metric values have been recorded in other SPI applications [90] including collaborative visioning for nuclear plant remediation [89]. In all these cases, the data gathered from participants demonstrates that satisfaction with the process is not linked to the level of satisfaction with the actual alternatives presented in the forums. That is, participants are able to distinguish between their judgments about the distributive justice implications of various alternations, and the procedural justice implications of how the decision process was being conducted [102]. In short, such processes deliver high EJ performance with respect to the DoT's second EJ aim [11]. Project management commentary supports this view e.g. [103]. This point is important, as it means that there is no good reason for public process professionals to avoid gathering Q-metric data as an initial basis for judging the quality of their processes.

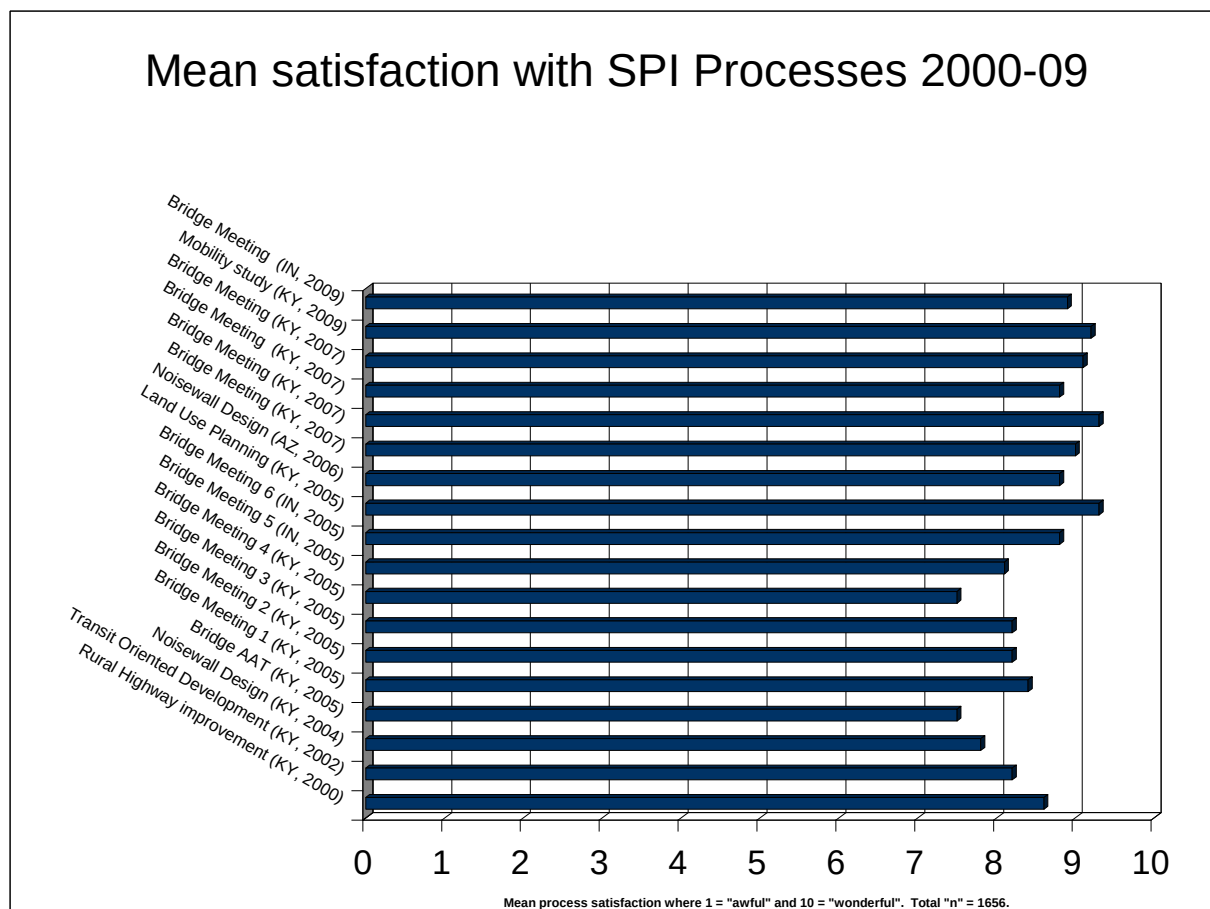


Figure 3. Q-metric data. Stakeholder Process performance Evaluation for SPI projects

Inclusion is an obvious metric, of interest to the EJ community to help ensure that project input is gathered from the broadest possible sectors of the overall community. For reasons we have discussed earlier, the lack of knowledge about suitable techniques for working with large groups, and resultant nervousness about the outcomes, often tends to drive practitioners toward multiple small groups, with attendant risks we have already discussed. Nevertheless, throughout the public involvement and EJ literature it is stressed that broader involvement is desirable. It is important to distinguish here between actual participation, comprised of useful feedback gathered, documented and incorporated, and measures of media contacts or other theoretical computations based on advertising exposure. We argue here for meaningful interaction, not marketing.

Clarity of decision support is directly related to the analytic sophistication of the methodology being used to convert the public process data into useful decision support for managers, engineers and designers. There is the question of decision making under uncertainty about public valuations. Often, public involvement data that is gathered piecemeal, or in an unstructured manner, is only of limited value under the best circumstances [63, p.4]. As an example, consultants often use visualizations of projects or designs as a method to force discrete choice. The injunction "pick one" or "vote for one" is used. But this approach can often create more problems than it solves. In experiments with real large-scale bridge design project, and in workshops delivered to DoT representatives, for example, the authors have compared ratio-scale

preference evaluation of the visualizations of design alternatives with forced choice, one-and-done voting. A comparison of the two methods, using exactly the same set of images, often yields different results. This finding is consistent with other environmental perception research [104,105]. Another case is the classic format two-hour meeting packed with consultants presenting information or data, which concludes with a five-minute session where “any comments” are solicited from public participants. These commentaries lack objective congruity with the decision objectives and are not easy to translate into meaningful design or planning guidance. In principle the information sought from the public must connect with, and it must also be seen to connect with, the choices available to the project team. And when the project conditions change to constrain the original options, these environmental changes can invalidate the data that has already been gathered. Deliberative democracy town hall approaches have also been criticized for their lack of integration into formal decision support for the clients [106].

Although public involvement is mandated, the cost of this undertaking is not often transparent. Typically this cost represents a very small fraction of overall project budget, despite entailing the significant risks of cost overrun or even project stoppage if not conducted at an acceptable quality threshold [107]. Even in straightforward cases, this cost is buried in project management spreadsheets. Over the life of projects, these costs are often not well tracked, particularly if further meetings are required as projects develop unexpectedly or as citizen opposition compels more public followup. Moreover, simply measuring the cost of public involvement is not helpful. A ratio of cost-effectiveness is required. Almost all other aspects of transportation projects are subject to a range of benefit-cost analyses, and yet the public involvement component is not subject to this same discipline. To operationalize an E-metric, other process performance metrics will need to be related to the cost of the undertakings.

IMPLICATIONS FOR ENVIRONMENTAL JUSTICE

The lack of performance evaluations of procedural justice is a serious problem for public involvement in transportation, just as it is for other public goods management fields. As Barnes and Langworthy [108, p.105] stated; “..there has been little attempt to develop [more general] theories within the context of transportation projects, possibly because systematic public involvement is a relatively recent development in this field.” A more rigorous and theoretical analytic approach to public involvement is required, one that frames the properties of the processes against the goals and that provides a framework for gathering of process performance data [109]. Some of this data must be gathered directly from stakeholders and citizens. Public involvement process design must address philosophical and theoretical considerations, as well as logistical issues such as the type of participation and the legal, institutional and social context within which the transportation planning process is situated. The methodology and logistics of SPI protocols are examples of processes designed to ensure the flattest possible distribution of power during meetings, and to respond to the metrics proposed earlier [90]. By so doing, such processes will better address issues facing the EJ community (and the community at large) [28].

A potential objection to over-reliance on a broad-reach approach is that special concerns of the more or less disenfranchised could be overwritten by those of majority groups. It is hard to produce factual evidence either in support of this view, or in contradiction. Some researchers

argue that this is already happening using the current methods [110]. Much depends on the geography within which the broader outreach is conducted [111]; for example, a series of open meetings conducted at locations determined to be of EJ significance could yield different results than a similar series with random distribution across the same enumeration unit [112]. A model of interest group politics in which those with greatest resources and access to power will dominate suggests that raising concerns in an adversarial, identifiable interest group framework will encourage or provoke factional conflict [113]. A precisely controlled, before-and-after experiment using the same subjects is not possible in a real project where participant learning is irreversible and where real decisions must be made on the basis of legally mandated stakeholder input.

In the methods field, research needs to be undertaken into large group process performance. At the same time, reliable, meaningful performance data needs to be defined and gathered in the course of real transportation projects using different methods. These data would allow for meaningful comparison between methods in specific situations. In the longer term these data sets would help create a body of knowledge that helps professionals and practitioners determine which processes are more suitable for which situations.

Such an endeavor may not always be welcomed by project sponsors or by contractors charged with implementing public involvement. Some Federal agencies are aware of internal resistance to such data-gathering efforts and quality metrics. For example, in the Champions of Participation report, internal inertia and opaque “risk-averse” behavior are identified as impediments within agencies to the acceptance and delivery of higher performance public involvement processes [47]. This is now beginning to be addressed; for instance, the proposal in this report that Federal officials in charge of public processes receive facilitation training is likely a helpful step [47]. So, too, is the opening of a discussion regarding potential quality metrics for public processes [47] including now from the TRB PI Committee [114].

Moreover, a philosophical shift is required, viewing the public as more than lip-service “partners,” but as responsible, interested parties who have meaningful information to contribute and whose values should be elicited and converted into decision support data in the most equitable, efficient and satisfying way possible. This entails developing integrated stakeholder-inclusive decision support systems that move all stakeholders – not only minority groups - towards the mutually-desired level of “partnership” (Level 6 on the Arnstein Ladder). It necessitates designing public involvement processes not to rely on any silencing or exclusion, but to foster the elicitation and capture of stakeholder valuations from the largest number of participants, in an efficient, equitable, and transparent manner, using technologies to accomplish these goals, and then to convert the data into decision support that allows effective alternatives analysis under both open and constrained conditions. These are not either/or, or zero-sum conditions.

Professional bodies including the Environmental Justice Committee of the Transportation Research Board and others have a key role to play in this conversation. In 2000, the TRB White Paper noted that: “The current mandates codify lessons learned in the 1970s and 1980s—lessons that many transportation agencies learned after the fact from project delays, lawsuits, and public outcry about transportation decisions made without citizen input.” [63, p.1]. To avoid this

reactive, inefficient and costly delayed response to previous mandates being repeated in the coming decade with respect to potential process quality metrics, it would be helpful to facilitate a more analytic discussion about public involvement methods, to inventory process metrics being used in other public process fields and data being gathered in the transportation field, and to convene fora to discuss this information. In this way, their EJ implications can be documented and evaluated by multiple investigators [94, 95]. Further, it seems an appropriate time to consider an updated White Paper on Public Involvement that includes a comprehensive and serious examination of potential performance metrics, a careful analysis of the role and merits of facilitation and electronic technologies in public processes, and linkages between these and EJ.

CONCLUSION

The transportation literature, and EJ literature more broadly, demonstrate that considerable effort is directed at training professionals to deliver small-scale processes that seek to identify minority groups and elicit their concerns. Identification is often based on sociological parameters such as income, ethnicity and education [115]. Even so, the methods often rely on participants' willingness to attend and speak in public forums, their ability to present arguments and their acceptance of their identity as defined by the project sponsor [116]. These assumptions are limiting. For instance, as Deka [117, p. 333] states: "Although these popular doctrines of social justice appear to be simple, it is difficult for many reasons to translate them objectively into public policies."

Following is but one example, from a document distributed in the transportation community as an example of exemplary group process (emphasis added):

"To convene the citizen advisory committee, VDOT placed advertisements in the city newspaper, as well as in smaller neighborhood newspapers. Nearly 70 people showed up at the first meeting. Overwhelmed with the response and feeling that the group would be unmanageable, VDOT asked the citizens to return to their respective interest organizations and identify only one or two persons who could serve for each organization. Individual citizens were told they could continue to participate as well. Significantly, while many different neighborhood and civic organizations were represented, more than one third of the people were also members of the same interest group, Friends of Bryan Park. Following VDOT's request, the group was thus pared down to about 30 citizen members composed of a mix of individuals and citizens representing a variety of local interests." [118]

Our goal here is not to condemn a particular process or agency. It is to illustrate how readily decisions are made that obviously compromise the rights of a majority of citizens, all in the name of practicality. In fact, they are held up as models of process. Conversely, effective large-group approaches avoid the situation described above, because they are designed for large groups: the more participants, the better. The meeting time demands are the same whether twenty people or three hundred are participating, but with more participants, the project sponsor acquires more robust data, allowing more defensible conclusions and recommendations. And when large groups are accommodated and encouraged, it is much more difficult for vested interest groups to challenge the "representative" quality of large-scale data sets on public values.

Unfortunately, large-group processes that seek to include stakeholders with widely varying levels of education, investment in the project at hand, time and other resources, are not well theorized, and metrics which can be used to evaluate real-world process performance are not widely documented [119]. Even the National Research Council's publications on public participation consider 'inclusiveness' to consist of "credible representatives of the full spectrum of parties" [120, p.230]. Nevertheless, according to Bullard and Johnson [121, p.555]; "it is possible to build a multiracial grassroots movement around environmental and economic justice." We might add, it may also build understanding among those with different incomes, education, employment patterns, and so on.

We believe this is also possible though inclusive, large-group processes oriented towards procedural justice and access to justice. Such processes can deliver low logistical, technical and educational barriers to participation, and higher procedural justice, with simultaneous, anonymous and independent stakeholder valuations that are visible to all in real time as a core component. We demonstrate that through reference to the example of multiple public processes conducted under the SPI protocol. The documented high Q-metric scores across multiple applications in different contexts and at different scales demonstrate that it is possible to systematically deliver high procedural justice performance in the absence of consensus and often in the absence of particular solutions that are favored. This data is helpful because it demonstrates that the public can distinguish between, and separately evaluate, the distributive justice implications of project options, and the procedural justice quality of the process itself, all in the same forum. Further, this can be done for all participants to a similar degree.

As we have argued earlier, the entire process of identification, segregation, and separation of EJ groups and issues from the larger community discussions is shot through with procedural justice issues that aggravate distributional justice issues. This does not mean that we advocate elimination of the conventional approach. Instead, we see value in significant augmentation of the EJ effort through a better articulation with a better-designed and delivered, more inclusive, high-performance public involvement strategy for projects. In terms of moving systemically towards stronger EJ across all three specified DoT aims, high-performance large group processes, as defined herein, can be implemented as a complementary methodology to current, smaller-scale, focused stakeholder engagement approaches. An integrated, multimethod approach can deliver enhanced equity in the form of broader inclusion, a flatter powerbase, efficiency for all participants in terms of limiting required meeting time, and access equity in terms of low participant training needs and modest resource demands. These methods can provide for ensuring better integration and understanding between project sponsors, consultants and the citizens at large within the ambit of the legal, the ethical, the financially and technically feasible. We hope the EJ community will investigate and document the performance of this, and other, comparative large-group processes in tandem with the traditional focused processes, so that more structured, data-rich comparative investigations of process performance with respect to EJ can be undertaken.

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